



Fig. 3: IDA-5 infusion pump tester and analyser (Credit: www.flukebiomedical.com)

ers are responsible for planning, performing and documenting the risk management process. They must estimate each applicable risk, probability of occurrence and severity of the risk both before and after risk mitigation measures have been applied. Each aspect of a risk management system should be thoroughly documented to control risk throughout the entire life of a given medical device.

Manufacturers must also ensure that their medical devices are free from bacteria and other substances that could transmit infectious diseases to patients and healthcare workers. This requires development, validation and routine control of sterilisation processes for medical devices and other healthcare products, as described in ISO 11135 for ethylene oxide, in ISO 11137 for gamma radiation and in ISO 17665 for moist heat.

Medical devices are subject to biological evaluation and biocompatibility testing to assess the interaction between a medical device or material that comes in contact with the patient's body. ISO 10993-1 specific testing is dependent on the type of medical device or material and its

intended use. It also depends on the nature and duration of contact between the medical device and the body.

Medical devices must undergo solid electronic testing to carry out testing of microprocessor chips before these are installed on a PCB. Tests for ICs consider their logic gate functions and interconnections between them. After attaching all components to the PCB, testing of mounting and interconnection process needs to be done.

Companies providing testing solutions for medical devices

Rigel Medical. Rigel Medical offers UniPulse 400, which is an easy-to-use defibrillator analyser with a pacer.

Compliant with IEC 60601-2-2, Uni-Therm electrosurgical analyser by Rigel Medical verifies and calibrates all major electrosurgical generators.

Rigel Medical's multi-flo infusion pump analyser meets the requirements of IEC 60601-2-24.

Eurofins Met Labs. This company provides a full range of electrical products as per 60601-1 series of standards. EMC and RFID susceptibility to verify compliance with major standards includes 60601, ISO 14708 and other international regulations. It is recognised by OSHA, which indicates compliance to standards required for workplace safety like hospitals, clinics, therapy centres and similar facilities.

Instron. It provides implantable device testing for contact lenses, heart pacemakers, ocular implants and stents. Testing systems are configured with BioBath 250 N submersible pneumatic grips for contact lens testing.

Instron 5900 series test systems with Bluehill Universal software and TestProfiler module are used for heart pacemaker testing.

Fluke Biomedical. It offers INCU II incubator tester to ensure quality

assurance and patient safety by complying with all global standards.

Its IDA-5 infusion pump tester and analyser provides quick preventive maintenance and repair with high accuracy.

Intertek. The company offers IEC 60601-1-2 compliance EMC testing, EMC edition 3 and EMC edition 4 testing for medical electrostatic discharge (ESD), EMC pre-compliance scans, radiated emissions, radiated immunity, MRI scanners, CT scanners and X-ray equipment, battery safety for medical devices to requirements of IEC 62133, and devices with non-rechargeable lithium batteries to IEC 60086-4.

Piper Medical. It provides a wide array of services for electrical and mechanical testing including biocompatibility, electrical safety, current leak, radiated immunity, EMI and ESD, mechanical reliability, environmental exposure, high-altitude, impact, shelf-life, vibration and IEC 60068-2.

Protomed Labs. It provides pulsatile flow pumps designed to create advanced cardiac or vascular flow loop systems for device testing. Its heart valve pulse duplicator system is designed to assess heart valve prosthesis performance under simulated cardiac conditions. Whereas, HiCycle determines the durability or fatigue of replacement heart valves and other cardiac devices under physiological loading for R&D and regulatory submissions.

To sum up

Testing of different functional blocks—such as operating system, instruments, hardware and so on—is validated for their intended use. It includes validation of specifications, testing and verification of performance of individual blocks to document results, which can be reviewed later.

While testing medical devices, accuracy within test results is crucial. Only high-quality testing in accordance with relevant standards can assure that products meet the specific requirements and do the job that these are designed for, in an approved and safe way. **EFY**



Fig. 4: Pulsatile flow pump (Credit: ViVibro Labs)